

TrueRisk

AI-Powered Multi-Hazard Risk Intelligence Platform

Hazard Intelligence Report

Generated 2026-07-02 | Coverage: Spain, 12 provinces | Forecast window: 30-day historical + 3-day outlook

This report summarizes the output of the TrueRisk pipeline: weather data ingestion, feature processing, CatBoost-based hazard prediction (heatwave, flood, wildfire), province-level risk scoring, and generated alerts with emergency guidance. Figures below are produced directly from the accompanying CSV outputs of this run.

1. Executive Summary

Across the 12 monitored provinces, 2 were classified HIGH risk and 0 CRITICAL at the time of this run. The highest overall risk score was recorded in **Valencia** at **56.7/100** (classification: HIGH). 2 active alerts were generated, each paired with hazard-specific emergency guidance.

2. Province Risk Scores

Province	Heatwave %	Flood %	Wildfire %	AQI %	Overall	Class
Valencia	96.9	35.8	29.7	41.0	56.7	HIGH
Alicante	93.3	53.6	6.1	43.0	56.4	HIGH
Toledo	96.3	4.1	12.3	42.5	43.8	MODERATE
Seville	94.1	6.6	13.3	39.5	43.5	MODERATE
Madrid	82.9	1.1	25.5	39.0	40.3	MODERATE
Barcelona	41.1	40.7	3.4	50.0	34.8	LOW
A Coruna	4.2	89.4	2.0	29.0	33.0	LOW
Malaga	46.1	24.5	3.6	34.0	29.3	LOW
Bilbao	5.8	70.7	1.4	32.0	28.3	LOW
Murcia	44.2	1.3	19.8	43.0	26.3	LOW
Granada	38.3	1.4	13.3	45.5	23.3	LOW
Zaragoza	36.3	3.0	8.8	42.5	21.7	LOW

3. Model Performance (Held-out Test Split)

Three CatBoost classifiers were trained — one per hazard type — on engineered weather features (temperature, humidity, rainfall, wind speed, AQI, coastal flag). Metrics below are computed on a 30% held-out test split, not the training data.

Hazard Model	Accuracy	Precision	Recall	F1 Score
heatwave	0.843	0.867	0.852	0.860
flood	0.852	0.818	0.730	0.771
wildfire	0.870	0.667	0.444	0.533
OVERALL (avg)	0.855	0.784	0.675	0.721

4. Active Alerts

Alert ID	Province	Hazard	Severity	Score
ALT-1000	Valencia	Heatwave	HIGH	56.7
ALT-1001	Alicante	Heatwave	HIGH	56.4

5. 3-Day Risk Outlook (Sample)

The table below shows the projected overall risk trajectory for the three highest-risk provinces.

Province	Day	Date	Predicted Risk	Class
Valencia	Day+1	2026-07-03	57.4	HIGH
Valencia	Day+2	2026-07-04	57.6	HIGH
Valencia	Day+3	2026-07-05	76.1	CRITICAL
Alicante	Day+1	2026-07-03	57.3	HIGH
Alicante	Day+2	2026-07-04	58.3	HIGH
Alicante	Day+3	2026-07-05	42.5	MODERATE
Toledo	Day+1	2026-07-03	42.9	MODERATE
Toledo	Day+2	2026-07-04	35.6	MODERATE
Toledo	Day+3	2026-07-05	37.1	MODERATE

6. Methodology & Data Notes

Pipeline stages: (1) weather/AQI ingestion per province, (2) feature engineering (heat index, dryness index, storm potential, rolling averages), (3) CatBoost classification per hazard type, (4) weighted aggregation into an overall 0-100 risk score (heatwave 35%, flood 30%, wildfire 20%, AQI 15%), (5) threshold-based alerting and guidance lookup, (6) 3-day forward projection.

Disclaimer: This report is generated from a demonstration dataset with synthetically constructed weather values and hazard labels for portfolio/showcase purposes. It illustrates the full pipeline architecture and modeling approach rather than live meteorological data or an operationally deployed early-warning system.